

Digital Transformation — Module 03: Perception —

Discussion Questions

20 questions on data analytics, organizational sensing, and digital perception.

1. How does analytics maturity (descriptive → predictive → prescriptive) correspond to increasingly sophisticated organizational inference?
2. Target could predict pregnancy from shopping data — but struggled to use this ethically. How should organizations navigate the data-to-decision gap?
3. When does more data improve organizational perception and when does it create noise that degrades perception?
4. How does real-time data change organizational decision-making compared to periodic reporting?
5. What are the most common data quality problems in organizations and how do they distort organizational inference?
6. When should an organization invest in prescriptive analytics versus improving descriptive analytics?
7. How does data bias (systematically distorted observations) differ from data insufficiency (too few observations)?
8. Compare the sensing capabilities of a data-rich organization versus a data-poor organization. What are the strategic implications?
9. How should privacy regulations (GDPR, CCPA) inform the design of organizational sensing systems?
10. When is qualitative data more valuable than quantitative data for organizational sensing?
11. Assess your organization's analytics maturity level. Where are the biggest gaps?
12. Map the data flows in your organization. Where does information get collected, processed, and used?
13. Design a data quality improvement program for your organization's most critical data asset.

14. Identify a decision in your organization that is currently made with inadequate data.
Design a sensing system to improve it.
15. How should organizations balance the cost of data collection with the value of the insights it produces?
16. Compare dashboard overload (too many metrics) with dashboard poverty (too few).
Which is worse? How to optimize?
17. How do A/B testing frameworks formalize the Active Inference perception process?
18. Design a "data governance" framework for an organization with sensitive customer data.
19. When should an organization build its own analytics capabilities versus buying from vendors?
20. How does the Internet of Things (IoT) expand organizational sensing to the physical world?